

THE RICE SIAM CHAPTER 50-DIGIT CHALLENGE

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Abstract. Recently the Rice University SIAM student chapter has published five new 10-digit problems. The intention is that Rice students solve every problem to as many digits of precision as possible. Archetype for this idea was a similar competition¹ which culminated in a beautiful book[1].

1. Introduction. Each of the questions below has an answer that is a single real number. The problems should be answered in the form $d_1.d_2\dots d_{10} \times 10^p$, $d_1 \neq 0$ (the p must be correct but does not count toward the number of correct digits), where d_{10} is determined by rounding to the nearest 10-digit real number. There are some further details but those apply only to Rice students.

1. Two players each control one knight at opposite corners on an 8×8 chessboard. They simultaneously try to move their respective pieces to an open square, using the L-shaped jump of the knight. Each of the valid jumps for either of the knights has equal probability of occurring. However if the two players choose to move their knights to the same square, they instead leave their pieces where they are and the turn is passed. What is the probability of at least one knight being at one of the four corners of the chessboard after 2005 turns?
2. A photon lies in the x - y plane trapped between two elliptical mirrors given by the equations $4x^2 + y^2 = 4$ and $9x^2 + 16y^2 = 144$. At time $t = 0$ the photon begins heading due north from position $(2, 0)$ with unit speed. How far is the photon from the origin at time $t = 60$?
3. Define the polynomial $p(x) = \sum_{k=0}^{10,000} a_k x^k$, where a_k is the $(k+1)$ st prime number. What is the magnitude of the root of p nearest the origin?
4. A point mass is at rest at the origin in the x - y plane. At time $t = 0$ the four other point masses listed in the table below are traveling clockwise with unit speed perpendicular to the radial direction. Each mass m_i attracts each other mass m_j with a gravitational force of magnitude $F_{ij} = m_i m_j / r^2$, where r is the distance of m_i from m_j . If these are the only forces present, how far from the origin is point p_4 at time $t = 5$?

Point	p_0	p_1	p_2	p_3	p_4
m_i	10	1	1	1	1
x_i	0	0	2	0	-4
y_i	0	1	0	-3	0

5. A sphere with unit radius centered at the origin in an (x, y, z) coordinate system is missing the part of its surface with z -coordinate greater than $3/4$. Two uniform balls with radius $1/4$ and unit mass are at rest at positions $(1/2, 1/8, 0)$ and $(0, 1/2, 0)$ inside the open “fishbowl.” At time $t = 0$, the balls are imparted with unit velocities toward the origin. What is the distance between the centers of the two balls at $t = 10$ if all collisions are completely elastic and there are no external forces?

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¹Nick Trefethen published 10 challenging problems. It’s already a tradition in Oxford that the new DPhil (PhD) students have to solve six problems in their first term.

Each of the next sections covers one problem. All results are explicitly listed in the conclusions at the end. I could not resist to include a suggestion for a sixth problem. In an Appendix I discuss the integral

$$\int_0^1 \sin^2(\tan(\tan(\pi x))) dx.$$

which is a problem I had some fun with.

2. A Markov Chain. We analyze the random walk of two knights on a chessboard.

PROBLEM. Two players each control one knight at opposite corners on an 8×8 chessboard. They simultaneously try to move their respective pieces to an open square, using the L-shaped jump of the knight. Each of the valid jumps for either of the knights has equal probability of occurring. However if the two players choose to move their knights to the same square, they instead leave their pieces where they are and the turn is passed. What is the probability of at least one knight being at one of the four corners of the chessboard after 2005 turns? $\diamond$

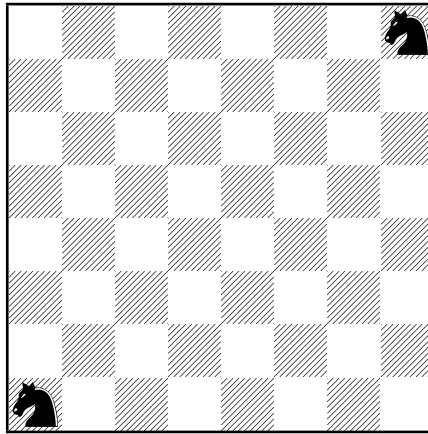


FIG. 2.1. *Initial position of the knights on the chessboard. The knight in the left corner is said to be in row 1 and column 1, whereas the other knight is in row 8 and column 8.*

To each field we assign a position p by $p_{ij} = 8(i - 1) + j$ where i is the row and j the column of the corresponding field. Hence the knights start at position 1 and 64. Knowing the position of both knights we can assign a unique number to any combination - we call those numbers the states s . Here we use $s = 64(p^{(1)} - 1) + p^{(2)}$ where $p^{(1)}$ and $p^{(2)}$ are the corresponding positions of knight 1 and knight 2. The initial position corresponds to the state $s = 64$. We distinguish $n = 64^2 = 4096$ states, although many states are not feasible. After every turn the knights are on fields of the same color. As the knights do not move in every turn we can not argue that they change color at every turn. Given a position p we compute the set of feasible new positions by:

```
function [newPos]=posMove(p);
```

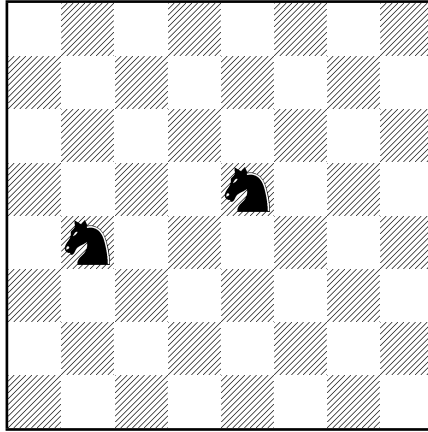


FIG. 2.2. A feasible state with $p^{(1)} = 26$ and $p^{(2)} = 37$ or vice versa. If both knights decide to move to position 20 or 43 respectively field d3 or c6 the turn is passed without move. For both players the set of feasible new fields is computed and all possible combinations are constructed.

```
[row,col]=posInv(p);

np=[row+2 col-1; row+2 col+1; row+1 col-2; row+1 col+2;...
    row-2 col-1; row-2 col+1; row-1 col-2; row-1 col+2];

np(np(:,1)>8,:)=[]; np(np(:,2)>8,:)=[];
np(np(:,1)<1,:)=[]; np(np(:,2)<1,:)=[];
newPos=pos(np(:,1),np(:,2));
```

Here is a list of all achievable fields from position 26 and 37. An example (compare with Figure 2.2):

```
>> x=sort(posMove(26)')
x =
     9    11    20    36    41    43
>> y=sort(posMove(37)')
y =
    20    22    27    31    43    47    52    54
```

Given the vectors of those fields the knights can achieve we construct the set of all possible combinations by using the Kronecker product:

```
function [p,m,n]=pairs(x,y);
% not that x and y might have different lengths
p=[kron(x,ones(size(y,1),1)) kron(ones(size(x,1),1),y)];
n=size(p,1);
p(p(:,1)==p(:,2),:)=[];
m=n-size(p,1);
```

Let us assume there are n pairs, where m of them are identical, i.e. they represent states that are not feasible as both knights try to move to the same field. For the example above we have $6 \times 8 = 48$ pairs, where two pairs are not feasible. If the knights are in state j and $i \neq j$ is a feasible new state then

$$p_{i,j} = \frac{1}{n}$$

is the probability to change from state j into state i , whereas

$$p_{j,j} = mp$$

is the probability to remain in state j . This defines a matrix $\mathbf{A}$.

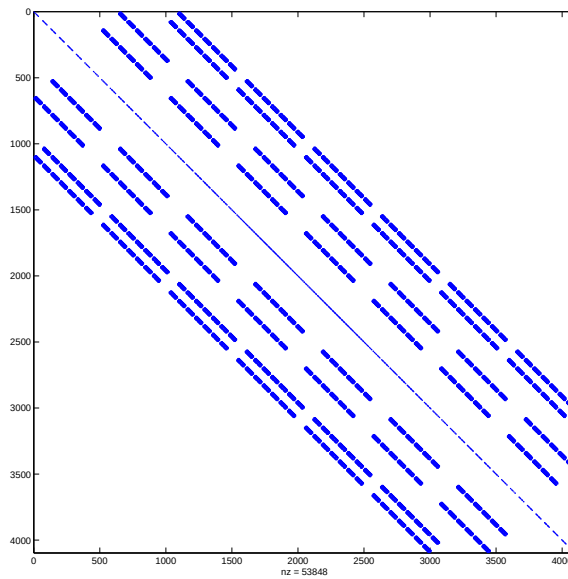
```
function [ A ] = TwoKnights
A=sparse(64^2,64^2);
for i=1:1:64
    for j=1:1:64
        if j~=i
            if color(i)==color(j)
                s=state([i j]);
                posI=posMove(i); posJ=posMove(j);
                % delete moves to occupied fields of the other knight
                posI(posI==j)=[]; posJ(posJ==i)=[];
                % bulid all possible pairs
                [p,m,n]=pairs(posI,posJ);
                prob=1/n;
                % Knights remain at the same place
                A(s,s)=m*prob;
                % Knights from state s to the states in p
                A(state(p),s)=prob;
            end
        end
    end
end
end
spy(A);
```

The matrix $\mathbf{A}$ is a 4096×4096 matrix with 53848 nonzero entries. All entries are nonnegative. If j is a feasible state then the sum of the entries of the j th column is 1. The initial state is 64 therefore we start with the vector $\mathbf{e}_{64} \in \mathbb{R}^{64^2}$ and compute the vector

$$\mathbf{v} = \mathbf{A}^{2005} \mathbf{e}_{64}$$

where the i th entry of $\mathbf{v}$ is the probability such that the two knights are in state i after 2005 moves. Let me emphasize that we do not compute $\mathbf{A}^{2005}$ and apply this on $\mathbf{e}_{64}$. It is faster to use a power iteration.

```
function [ v ] = powerIteration( A,v,m )
for i=1:1:m
    v=A*v;
end
```

FIG. 2.3. *Spy plot of the 4096×4096 matrix $\mathbf{A}$.*

Putting everything together we end up with

```
tic
A=TwoKnights;
toc

tic
v=zeros(4096,1); w=v';
s0=state([1 64]); v(s0,1)=1;
v=powerIteration(A,v,2005);
s=sum(v)
corner=[1 8 57 64]'; all=(1:1:64)';
states=state([pairs(corner, all); pairs(all,corner)]);
w(1,states)=1;
p=w*v
toc
```

We construct all pairs and the corresponding states where at least one knight is in a corner. Then j th entry of $\mathbf{v}$ is then the probability of the system being in state j . Adding all those entries we are done. In order to check the accuracy we add all entries of $\mathbf{v}$. This sum should be 1. Here are our results

Elapsed time is 1.102000 seconds.

```
s =
    0.999999999999998
```

```
p =
    0.04849375459221
Elapsed time is 2.504000 seconds.
```

The matrix $\mathbf{A}$ has a remarkable property. It has an eigenvalue that is exactly 1 whereas the other eigenvalues lie within a disk around the origin with radius smaller than 0.95. The matrix $\mathbf{A}$ is essentially a stochastic column matrix, that is for every column the sum of its entries is 1 and there are no negative entries. Note that we could get rid of all columns and rows with no entries, such that this is fulfilled. We denote the dominating eigenvector by $\mathbf{u}_1$. As $\mathbf{e}_{64}^H \mathbf{u}_1 \neq 0$ the vector $\mathbf{v}$ will point into the direction of $\mathbf{u}_1$. The length of $\mathbf{v}$ is fixed by the constraint that $\|\mathbf{v}\|_1 = 1$. Therefore we get

$$p = \frac{\mathbf{w}^H \mathbf{u}_1}{\|\mathbf{u}_1\|_1}.$$

where $\mathbf{w}$ is again the vector that counts the entries corresponding to states where at least one knight is in a corner.

```
tic
opts disp=0
[u,D]=eigs(A,5,'LM',opts);
D=diag(D)
u=u(:,1);
p=abs(w*u/norm(u,1))
toc
```

The result results perfectly fit those results gained before

```
D =
    1.000000000000000
   -0.94309049561796
    0.74020698713032
    0.74020698713032
    0.72389482491549
p =
    0.04849375459221
Elapsed time is 0.521000 seconds.
```

Hence I submit as solution for this problem

0.04849375459.

3. A Photon trap. We chase the path of a photon trapped between two elliptic mirrors.

PROBLEM. A photon lies in the x - y plane trapped between two elliptical mirrors given by the equations $4x^2 + y^2 = 4$ and $9x^2 + 16y^2 = 144$. At time $t = 0$ the photon begins heading due north from position $(2, 0)$ with unit speed. How far is the photon from the origin at time $t = 60$? $\diamond$

The best start is probably a plot of both elliptical mirrors. Here we denote by a the length of the semiaxis in x -direction and by b the length of the semiaxis in

y -direction. We use the terms inner and outer ellipse. They mean exactly what you think they mean. For the inner ellipse we have $a = 1$ and $b = 2$, whereas for the outer ellipse we have $a = 4$ and $b = 3$. A moving photon was also an issue in the original

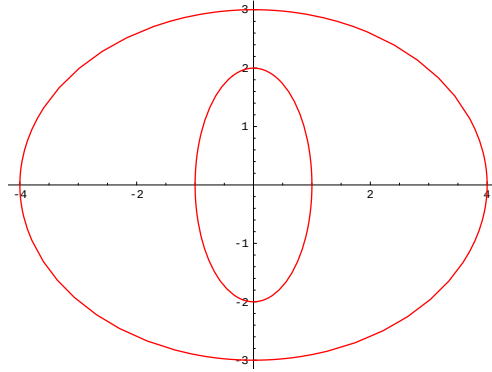


FIG. 3.1. *The two elliptical mirrors*

competition. Wagon discusses the problem in a chapter of [1]. A photon trapped between the two mirrors at position $\mathbf{p}$ with velocity $\mathbf{v}$ and without loss of generality starting at time $t_0 = 0$ will hit the outer mirror at time t_1 where t_1 is the positive parameter such that the point

$$\mathbf{p} + t_1 \mathbf{v}$$

lies on the ellipse. In MATHEMATICA we use:

```
Ndigit = 100;
p = N[{2, 0}, Ndigit]; v = N[{0, 1}, Ndigit]; tRem = 60;
t1 = Min[Cases[t/.
    Solve[ 9*(p[[1]]+t*v[[1]])^2+
          16*(p[[2]]+t*v[[2]])^2==144,t],_?Positive]];
```

However, before the photon might collide with the inner mirror. If the photon is not moving in a direction that leads to a collision with the this mirror the corresponding parameter for the time t_2 is set to ∞ .

```
t2 = Min[Cases[t/.
    Solve[ 4*(p[[1]]+t*v[[1]])^2+
          (p[[2]]+t*v[[2]])^2==4,t],_?Positive]];
```

The photon will not hit any mirror if the remaining time t_r is smaller than t_1 and t_2 .

```
deltaT = Min[{t1, t2, tRem}];
newpos = pos + deltaT*v;
```

Hence the photon will move from position $\mathbf{p}$ to position $\mathbf{p}' = \mathbf{p} + \Delta T \mathbf{v}$ where $\Delta T = \min\{t_1, t_2, t_r\}$. If the new position

$$\mathbf{p}' = \begin{pmatrix} x \\ y \end{pmatrix}$$

is not the final state of the photon it will be reflected at the corresponding mirror. The photon is approaching $\mathbf{p}'$ with the velocity $\mathbf{v}$. The unit tangent vector at $\mathbf{p}'$ is

$$\begin{pmatrix} x_T \\ y_T \end{pmatrix} = \frac{1}{\|\mathbf{p}'\|_2} \begin{pmatrix} -a/b y \\ +b/a x \end{pmatrix}$$

where a and b are the semiaxes of the corresponding ellipse and the outward unit normal vector is

$$\begin{pmatrix} y_T \\ -x_T \end{pmatrix}.$$

The reflection of the velocity vector is a linear map, that keeps the unit tangent vector invariant. The difference between the two ellipses is the reflection of the unit normal vector. If $\mathbf{v}$ is the outward unit normal vector and the photon hits the outer ellipse then $\mathbf{v}'$ is the inward unit normal vector. If $\mathbf{v}$ is the inward normal vector and the photon hits the inner ellipse then $\mathbf{v}'$ is the outward unit normal vector. This defines two reflection matrices $\mathbf{R}_1$ for the inner ellipse and $\mathbf{R}_2$ for the outer ellipse.

$$\mathbf{R}_1 \begin{pmatrix} -y_T & x_T \\ x_T & y_T \end{pmatrix} = \begin{pmatrix} y_T & x_T \\ -x_T & y_T \end{pmatrix}$$

and

$$\mathbf{R}_2 \begin{pmatrix} y_T & x_T \\ -x_T & y_T \end{pmatrix} = \begin{pmatrix} -y_T & x_T \\ x_T & y_T \end{pmatrix}$$

Please note that

$$\mathbf{R}_2^{-1} = \mathbf{R}_1$$

In MATHEMATICA this reads

```
T[{p_,a_,b_}]:= (1/Norm[p,2] )*{-a*p[[2]]/b,b*p[[1]]/a};
R1[{x_,y_}]:= {{y,x},{-x,y}}.Inverse[{{-y,x},{x,y}}];
R2[{x_,y_}]:=Inverse[R1[{x,y}]];
```

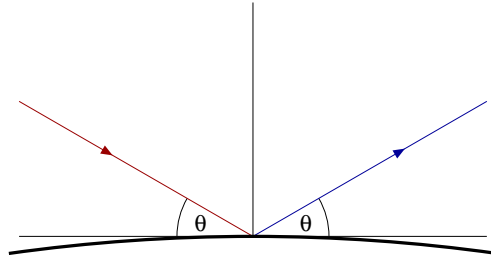


FIG. 3.2. *Reflection at a curved boundary*

The idea is to define a function that maps the old position and velocity vector to the new data.

```

hit[{pos_,v_,tRem_}]:=Module[{},
  t1=Min[
    Cases[t/.Solve[
      9*(p[[1]]+t*v[[1]])^2+16*(p[[2]]+t*v[[2]])^2==144,t],
      _?Positive]];
  t2=Min[
    Cases[t/.Solve[
      4*(p[[1]]+t*v[[1]])^2+(p[[2]]+t*v[[2]])^2==4,t],
      _?Positive]];
  deltaT=Min[{t1,t2,tRem}];
  newpos=pos+deltaT*v;
  If[deltaT==t1,newv=R2[T[{newpos,4,3}]] .v];
  If[deltaT==t2,newv=R1[T[{newpos,1,2}]] .v];
  {newpos,newv,tRem-deltaT}
]

```

Then we iterate as long as there is some time remaining.

```

While[tRem>0,
  {p,v,tRem}=hit[{p,v,tRem}];setP=Append[setP,p];

```

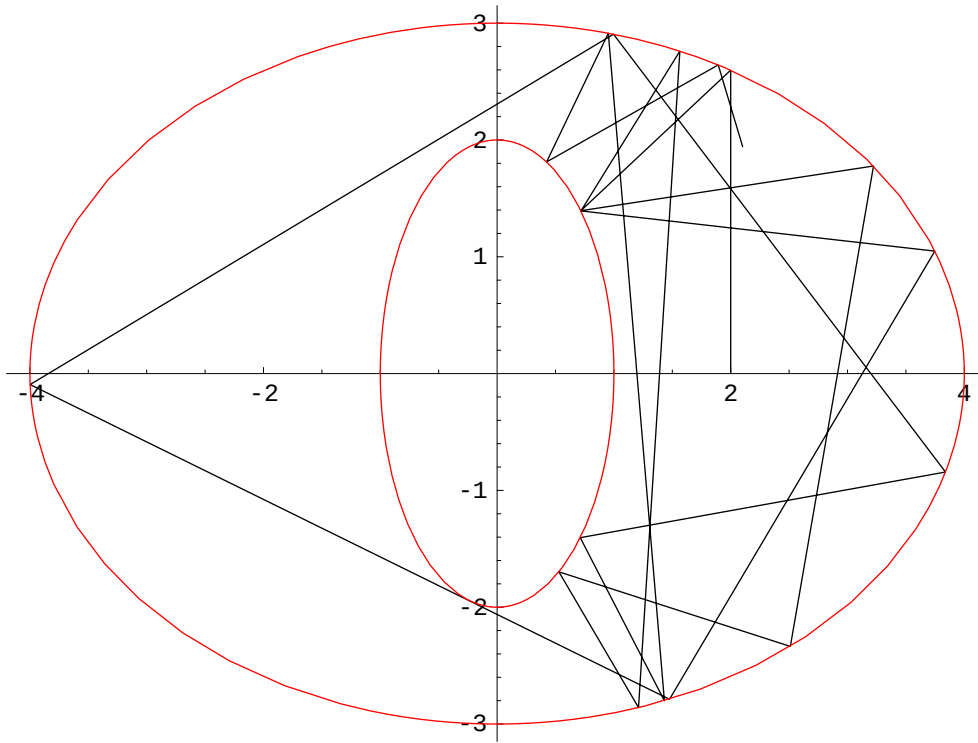


FIG. 3.3. The path of the photon between the two mirrors

Wagon [1] points out, that reflections are very sensitive to variation in the initial

data. Reflections are ill-conditioned. We can show this effect by varying the number of digits MATHEMATICA uses in all operations. Performing elementary operations MATHEMATICA cuts off digits that are no reliable. Here is a table

Number of digits	final distance from the origin	Digits in the result
40	3.3	2
41	2.86	3
42	2.862	4
45	2.8623	5
50	2.86233587	9
60	2.8623358681628731	17
70	2.86233586816287309621536776	27

It turns out that a loss of 43 digits is unavoidable. In practice it is possible to increase the number of digits used almost arbitrarily. It is no problem at all to get 10000 digits.

I submit as solution for this problem

$$2.862335868.$$

4. An eigenvalue problem. We compute the root closest to the origin of a polynomial of degree $n = 10000$.

PROBLEM. Define the polynomial $p(x) = \sum_{k=0}^{10,000} a_k x^k$, where a_k is the $(k+1)$ st prime number. What is the magnitude of the root of p nearest the origin? $\diamond$

The roots of a monic polynomial

$$p(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$$

are the eigenvalues of the *companion matrix*

$$\mathbf{A} = \begin{pmatrix} -a_{n-1} & -a_{n-2} & \dots & \dots & -a_1 & -a_0 \\ 1 & & & & & \\ & 1 & & & & \\ & & \ddots & & & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & 1 \end{pmatrix}$$

In MATLAB we can exploit the sparsity structure of this matrix:

```
function [ A ] = spcompan( a )

a=a/a(end);      % monic polynomial
a=-a(2:end);     % the negative coefficients in the first row
k=length(a);
A=[fliplr(a); speye(k-1,k)];
```

We can scale any polynomial by dividing with the leading coefficient such that it is monic. Here it is

$$a_{10000} = 104743.$$

Instead of using a polynomial of degree 10000 we cut off all orders higher than a constant M . Hence $p^M(x) := \sum_{k=0}^M a_k x^k$. The triangle inequality gives the simple estimation

$$|p(x)| \leq |p^M(x)| + 10000 \times a_{10000} |x|^M \quad \forall |x| < 1.$$

The second addend is small if x is well separated from the unit sphere and M is reasonable large. Let us first restrict our search for to the ball around the origin with radius 0.82. Let $M = 300$ then

$$10000 \times a_{10000} |x|^M < 1.5 \times 10^{-17} \quad \forall |x| < 0.82.$$

Hence instead of minimizing $|p(x)|$ we minimize $|p^{300}(x)|$ which is simpler and faster, since the roots of $p^{300}(x)$ are the eigenvalues of a 300×300 companion matrix associated with this polynomial. The computation of those eigenvalues is by the two standard approaches - the QR algorithm and the Arnoldi method. The interested reader should consult the textbook by Trefethen and Bau [2].

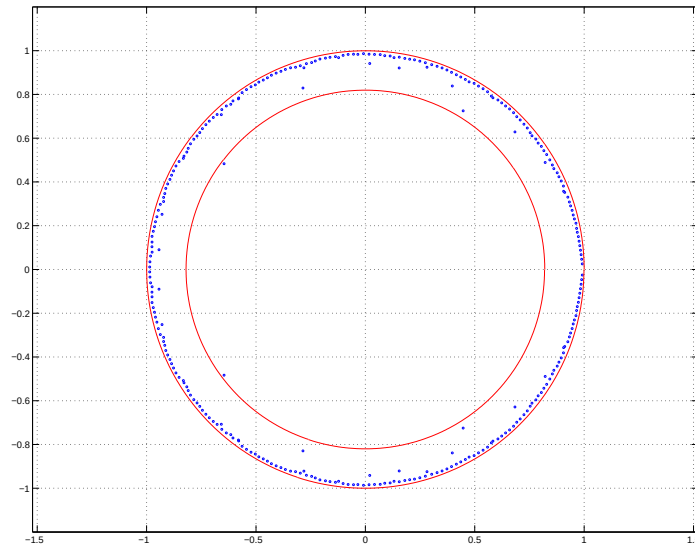


FIG. 4.1. The spectrum of the companion matrix of $p^{300}(x)$. The circles have radii 1 and 0.82.

The companion matrix $\mathbf{A}$ is constructed here:

```
M=300;
% a is a vector of all primes smaller than 1e6
a=primes(1e6);
a=a(1:1:M+1);
A=spcompan(a);
```

The QR algorithm is not designed for sparse matrices.

```
tic
D=eig(full(A));
v1=min(abs(D))
toc
```

Here we basically use an inverse power iteration without shifts.

```
tic
opts.tol=1e-15; opts.disp=0;
v2=abs(eigs(A,1,'SM',opts))
toc
```

Both results agree at least to ten digits

```
v1 =
    0.80651359926038
```

```
Elapsed time is 0.340000 seconds.
v2 =
    0.80651359926048
```

```
Elapsed time is 0.161000 seconds.
Hence I submit for this problem
```

0.8065135993.

5. A five-body problem. The paths of five planets in a flat universe are determined by numerical integration.

PROBLEM. A point mass is at rest at the origin in the x - y plane. At time $t = 0$ the four other point masses listed in the table below are traveling clockwise with unit speed perpendicular to the radial direction. Each mass m_i attracts each other mass m_j with a gravitational force of magnitude $F_{ij} = m_i m_j / r^2$, where r is the distance of m_i from m_j . If these are the only forces present, how far from the origin is point p_4 at time $t = 5$? $\diamond$

Point	p_0	p_1	p_2	p_3	p_4
m_i	10	1	1	1	1
x_i	0	0	2	0	-4
y_i	0	1	0	-3	0

We identify the phase space as $\mathbb{C}^{10}$ as follows: Let $(\mathbf{u}, \mathbf{v})^\top$ be a vector in phase space where $\mathbf{u} \in \mathbb{C}^5$ is the position of the planets using complex coordinates and $\mathbf{v} \in \mathbb{C}^5$ is the corresponding velocity. For this decomposition we have

$$\frac{d}{dt} \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix} = \begin{pmatrix} \mathbf{v} \\ \mathbf{a} \end{pmatrix} \quad (5.1)$$

where $\mathbf{a} \in \mathbb{C}^5$ is the acceleration. Newtons second law states

$$\mathbf{F}_i = m_i \mathbf{a}_i \quad \forall i = 1, \dots, 5$$

where $\mathbf{F}_i$ is the force acting on the i th planet. We have

$$\mathbf{F}_i = \sum_{\substack{j=1 \\ j \neq i}}^5 \mathbf{F}_{ij} = m_i \sum_{\substack{j=1 \\ j \neq i}}^5 m_j \frac{\mathbf{u}_i - \mathbf{u}_j}{|\mathbf{u}_i - \mathbf{u}_j|^3},$$

and hence

$$\mathbf{a}_i = \sum_{\substack{j=1 \\ j \neq i}}^5 m_j \frac{\mathbf{u}_i - \mathbf{u}_j}{|\mathbf{u}_i - \mathbf{u}_j|^3}.$$

Equation (5.1) is in MATLAB:

```
function du = planetsfun(t,u,m)

du=zeros(10,1);
du(1:5)=u(6:10);
force=zeros(5,1);
for j=1:1:5;
    % compute force on particle j
    for k=1:1:5
        if k==j
            force(k)=0;
        else
            force(k)=(u(k)-u(j))/abs(u(k)-u(j))^3;
        end
    end
    du(j+5)=m*force;
end
```

We integrate the ordinary differential equation (5.1) by standard solvers implemented in MATLAB:

```
tic
% Initial conditions:
% mass:
m=[10 1 1 1 1];
% position in the phase space C^10
u0=[0 i 2 -3i -4 0 1 -i -1 i].';
opts = odeset('reltol',2.22045e-014,'abstol',1e-20);

% Time-stepping:
[t,u] = ode113(@planetsfun,[0,5],u0,opts,m);
u0 = u(end,:);
val=abs(u0(5))
toc
```

The program terminates with a value of

$$1.89450979564148.$$

How can we guarantee that at least ten digits of this result are correct? The idea is to perturb the initial conditions and to repeat the integration process

```
wEPS=1
% Initial conditions:
if wEPS
    epsilon=rand(1)*1e-15+i*rand(1)*1e-15
else
    epsilon=0
end
% position in the phase space C^10
u0=[0 i 2 -3i -4+epsilon 0 1 -i -1 i].';
```

We measure at which scale tiny perturbations of the initial position of the 5th planet are reflected (emphasized).

Perturbation	final distance
$1.527392702903627e - 017 + 7.467856765644294e - 016i$	1.89450979565940
$9.218129707448026e - 016 + 7.382072458106653e - 016i$	1.89450979587871
$1.762661444946180e - 016 + 4.057062130620955e - 016i$	1.89450979562254
$9.354696991076055e - 016 + 9.169044399134077e - 016i$	1.89450979566157
$4.102702069909454e - 016 + 8.936495309135336e - 016i$	1.89450979583813
$5.789130478426856e - 017 + 3.528681322170004e - 016i$	1.89450979562855
$8.131664973037578e - 016 + 9.861300660923561e - 018i$	1.89450979563701
$1.388908819569499e - 016 + 2.027652185602732e - 016i$	1.89450979558832
$1.987217426614897e - 016 + 6.037924791938193e - 016i$	1.89450979582276
$2.721879249699604e - 016 + 1.988142677610622e - 016i$	1.89450979565492

Note that the only the first ten digits are equal for every experiment. From this experiment we conclude that the solution is

$$1.894509796.$$

Unfortunately we can not give additional digits for this problem.

6. Balls in the fishbowl. Two balls are bouncing around in a fishbowl.

PROBLEM. A sphere with unit radius centered at the origin in an (x, y, z) coordinate system is missing the part of its surface with z -coordinate greater than $3/4$. Two uniform balls with radius $1/4$ and unit mass are at rest at positions $(1/2, 1/8, 0)$ and $(0, 1/2, 0)$ inside the open “fishbowl.” At time $t = 0$, the balls are imparted with unit velocities toward the origin. What is the distance between the centers of the two balls at $t = 10$ if all collisions are completely elastic and there are no external forces? $\diamond$

This problem is closely related to the problem of the photon trapped between the mirrors. For the solution we have recycled the code before. However, here we have two bodies involved. Although the problem is stated in three dimensions it is essentially a two dimensional problem as the midpoint of both balls remain in the layer with $z = 0$.

With $\mathbf{p}_1$ and $\mathbf{p}_2$ we denote the position of the midpoint of both disks. The corresponding velocities are $\mathbf{v}_1$ and $\mathbf{v}_2$. When it comes to a collision energy and the momentum are conserved. The center of mass is the midpoint of line connecting both midpoints of the discs. We denote its position by $\mathbf{p}'$. It is

$$\mathbf{p}' = \frac{\mathbf{p}_1 + \mathbf{p}_2}{2}.$$

The center of mass moves with velocity

$$\mathbf{v}' = \frac{\mathbf{v}_1 + \mathbf{v}_2}{2}.$$

Assume we observe the collision from the center of mass. In this system the collision is very easy to describe as both discs approach the center of mass from contrary directions. The approach the center of mass with velocity

$$\mathbf{v}'_1 = \mathbf{v}_1 - \mathbf{v}'$$

and

$$\mathbf{v}'_2 = -\mathbf{v}'_1.$$

Therefore

$$\mathbf{v}_1 = \mathbf{v}' + \mathbf{v}'_1$$

and

$$\mathbf{v}_2 = \mathbf{v}' - \mathbf{v}'_1.$$

The components $\mathbf{v}'_1$ and $\mathbf{v}'_2$ change their sign therefore we end up with the simple observation that the discs swap their velocity vectors when it comes to a collision.

```
<<Graphics'Graphics'
<<Graphics'Colors'
```

```
Ndigit=100;
```

```
R[{x_,y_}]:={{-x,y},{-y,-x}}.Inverse[{{x,y},{y,-x}}];
```

```
r=N[1/4,Ndigit];
```

```
p1=N[{1/2,1/8},Ndigit]; p2=N[{0, 1/2},Ndigit];
```

```
v1=-p1/Norm[p1,2];v2=-p2/Norm[p2,2];
```

```
tRem=10;
```

```
setP1={p1};setP2={p2};
```

```
hit[{p1_,p2_,v1_,v2_,tRem_}]:=Module[{},
  r1=Min[Cases[t/.Solve[(p1+t*v1).(p1+t*v1)==(1-r)^2,t],_?Positive]];
  r2=Min[Cases[t/.Solve[(p2+t*v2).(p2+t*v2)==(1-r)^2,t],_?Positive]];
  r3=Min[Cases[t/.Solve[(p1-p2+t*(v1-v2)).(p1-p2+t*(v1-v2))==4*r^2,t],
    _?Positive]];
  r4=tRem;
```

```

l=Min[{r1,r2,r3,r4}];
newp1=p1+l*v1;
newp2=p2+l*v2;
If[l==r1, newv1=R[newp1].v1];
If[l==r2, newv2=R[newp2].v2];
If[l==r3, newv1=v2; newv2=v1];
{newp1,newp2,newv1/Norm[newv1,2],newv2/Norm[newv2,2],tRem-1,1}
]

While[tRem>0,
  {p1,p2,v1,v2,tRem,l}=hit[{p1,p2,v1,v2,tRem}];
  setP1=Append[setP1,p1]; setP2=Append[setP2,p2]];

```

It might be difficult to understand the code above. However, I refer the reader to the second problem. Here we have 4 different scenarios. Ball 1 can hit the boundary, ball 2 can hit the boundary, ball 1 can hit ball 2 or there is no remaining time left.

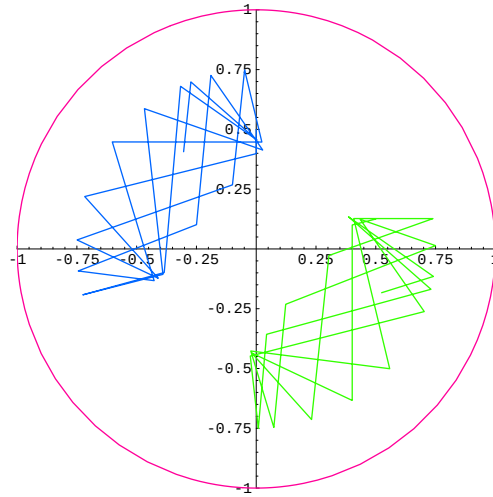


FIG. 6.1. The path of the midpoints of both balls at time $t = 10$.

As answer for this problem I submit

1.016560194

7. Conclusions. The problems provided some fun. Although I think that the problems 2 and 5 are too closely related. An alternative problem is stated in the Appendix. Here are results again

Problem	Result
1	0.04849375459
2	2.862335868
3	0.8065135993
4	1.894509796
5	1.016560194

The numbers come without guarantee.

REFERENCES

- [1] F. BORNEMANN, D. LAURIE, S. WAGON AND J. WALDVOGEL, *The SIAM 100-Digit Challenge*, SIAM, Philadelphia, 2004.
- [2] L. N. TREFETHEN AND D. BAU, III, *Numerical Linear Algebra*, SIAM, Philadelphia, 1997.

Appendix. There might be a tie in the end of the competition. For this case there will be an additional problem published. Here is my favorite problem and suggestion for this case.

PROBLEM. Compute

$$\int_0^1 \sin^2(\tan(\tan(\pi x))) dx \quad (\text{A.1})$$

◇

It was Brian Davies who pointed out that the evaluation would involve serious difficulties both numerically and analytically. Indeed it is hard to integrate the function on the real line but a small step to the complex plane will give us some insight.

A.1. Existence. The function

$$f(x) = \sin^2(\tan(\tan(\pi x)))$$

is not defined for some $x \in [0, 1]$. The $\tan$ -function has poles at $\frac{\pi}{2} + k\pi$ for all $k \in \mathbb{Z}$. Obviously f is therefore not defined for all elements of the countable set

$$X = \left\{ \frac{1}{\pi} \arctan\left(\frac{\pi}{2} + k\pi\right), k \in \mathbb{Z} \right\} \cup \left\{ \frac{1}{2} \right\}.$$

Estimating $0 \leq f(x) \leq 1$ for all $x \in [0, 1] \setminus X$ implies that f is integrable and

$$0 \leq \int_0^1 f(x) dx \leq 1.$$

A.2. Trigonometric functions in the complex plane. It is

$$e^{iz} = \cos z + i \sin z$$

and

$$e^{-iz} = \cos z - i \sin z.$$

Hence for $z \in \mathbb{C}$ it is

$$\sin^2 z = \left(\frac{1}{2i} (e^{iz} - e^{-iz}) \right)^2 = \frac{1}{2} - \frac{1}{4} (e^{2iz} + e^{-2iz}) = \frac{1}{2} - \frac{1}{2} \cos(2z)$$

For $x \in \mathbb{R}$ this is

$$\sin^2 x = \frac{1}{2} - \frac{1}{2} \Re(e^{i2x}).$$

or as $\Re z = \Re \bar{z}$

$$\sin^2 x = \frac{1}{2} - \frac{1}{2} \Re(e^{-i2x}).$$

Both identities can be continued to the complex plane. But note that they are equal to $\sin^2 z$ only on the real line. The function e^{i2z} is bounded in the upper halfplane, as $|e^{i2z}| \leq 1$ for all z with $\Im z \geq 0$. The function e^{-i2z} is bounded in the lower halfplane.

It is now possible to rewrite to integral as

$$\frac{1}{2} - \frac{1}{2} \Re \int_0^1 e^{\pm 2i \tan(\tan(\pi x))} dx$$

Let $c \geq 0$ an arbitrary constant. We introduce the notation

$$f_c^+(x) = e^{+2i \tan(\tan(\pi x + ic))}$$

and

$$f_c^-(x) = e^{-2i \tan(\tan(\pi x - ic))}$$

LEMMA A.1. *The complex tan-function maps the upper halfplane to the upper halfplane and the lower halfplane to the lower halfplane.*

Proof. Let $z = x + iy$. It is

$$\tan z = \frac{\sin z}{\cos z} = \frac{\sin z \overline{\cos z}}{|\cos z|^2}.$$

But

$$\sin z = \sin x \cosh y + i \cos x \sinh y \quad \cos z = \cos x \cosh y - i \sin x \sinh y$$

implies

$$\Im \tan z = \sinh y \underbrace{\frac{\cosh y}{|\cos z|^2}}_{>0 \ \forall z \in \mathbb{C}}.$$

But $\sinh y > 0$ if and only if $y > 0$ and $\sinh y < 0$ if and only if $y < 0$. $\square$

As a consequence we note that $|f_c^+| \leq 1$ for all $c > 0$.

A.3. Dominated convergence. For positive c the functions f_c^+ are integrable and $f_c^+ \xrightarrow{c \rightarrow 0} e^{+2i \tan(\tan(\pi x))}$ almost everywhere, that is for all $x \in [0, 1] \setminus X$. The theorem of dominated convergence yields as $|f_c^+| \leq 1$

$$\int_0^1 f_c^+ dx \xrightarrow{c \rightarrow 0} \int_0^1 e^{+2i \tan(\tan(\pi x))} dx. \quad (\text{A.2})$$

Using the same argument

$$\int_0^1 f_c^- dx \xrightarrow{c \rightarrow 0} \int_0^1 e^{-2i \tan(\tan(\pi x))} dx.$$

A.4. Cauchy's Theorem. The function f_0^+ is analytic in the open upper half-plane. By Cauchy's theorem, the line integral between two points in the complex plane does not depend on the path taken, as long as both paths lie in a region inside which the function is analytic. As path we use the edges A, B, C and D of a rectangle in the complex plane. Cauchy's theorem implies

$$\int_A f_0^+ + \int_B f_0^+ + \int_C f_0^+ + \int_D f_0^+ = 0$$

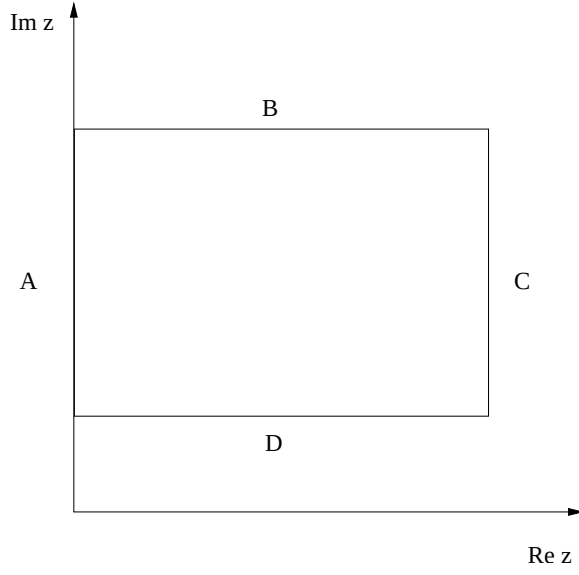


FIG. A.1. The path of integration in the complex plane.

If $1 + ic_1$ and $1 + ic_2$ are the endpoints of edge C we observe due to the periodicity of the function f_0^+ :

$$\int_A f_0^+ + \int_C f_0^+ = 0.$$

The definition of the path integral yields

$$\int_B f_0^+ = \int_0^1 f_{c_1}^+ dx = \int_0^1 f_{c_2}^+ dx = \int_D f_0^+.$$

As the constants c_1 and c_2 are arbitrary the integral does not depend on the choice of the positive bias c .

$$\int_0^1 f_c^+ = C \quad \forall c > 0.$$

Taking the limit (A.2) implies

$$\int_0^1 e^{+2i \tan(\tan(\pi x))} dx = C$$

Keeping in mind that the original function f_0^+ on the real line is rather nasty it is surprising that the functions f_c^+ are converging pointwise to the most simple thinkable function - a constant function. Hence the constant C is given by

$$\lim_{c \rightarrow \infty} e^{2i \tan(\tan(\pi x + c i))} = e^{2i \tan(i)} = e^{-2 \tanh(1)}.$$

Putting the results together we get

$$\int_0^1 \sin^2(\tan(\tan(\pi x))) dx = \frac{1}{2} - \frac{1}{2} e^{-2 \tanh(1)} = 0.39099216215153 \dots$$

The same argument can be applied to f_c^- in the lower halfplane.